Conduction, convection, radiation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify mechanisms of heat transfer and support each classification.

SUCCESS CRITERIA - I can...

1. Describe contact transfer, bulk fluid motion and electromagnetic transfer.
2. Use evidence from the situation.
3. Recognize that more than one mechanism can act.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A device transfers 15 J in 3 s, then performs the same transfer in 6 s. Find both average powers and explain the difference.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $E_i + E_{in} = E_f + E_{out}$.
Account for energy across the system boundary. Conduction: matter interactions; convection: fluid motion; radiation: electromagnetic waves.

Worked reasoning: A metal rod transfers energy from its hot end toward its cool end by conduction. A circulating fluid can carry energy by convection; radiation can cross a vacuum.

Your turn: Draw one distinct, labeled model for each mechanism. Show the direction of energy transfer.

Marzano check - circle: ① ② ③

03 Look and explain

Classify: a metal spoon warming in soup; circulating water in a pot; sunlight warming your hand. Give a reason for each.

Marzano check - circle: ① ② ③

04 Build the solution

Draw arrows showing warm fluid rising and cooler fluid sinking in a simple convection loop. Explain why this is not the same as conduction along a solid spoon.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

05 Use the idea

An asteroid travels through nearly empty space. Which of these mechanisms can transfer thermal energy to it from sunlight? Why not ordinary air convection?

Marzano check - circle: 1 2 3

06 Choose a method

A rolling object loses 24 J gravitational energy but gains only 16 J translational kinetic energy. Name two possible destinations of the difference.

Marzano check - circle: 1 2 3

07 Exit check

Explain how energy can travel from the Sun to Earth through space.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Design insulation for a hot cup using one change for each transfer mechanism. Explain each change.

Marzano check - circle: 1 2 3